

International Cybersecurity and Digital Forensics Academy (ICDFA)

CIP-B102 Foundations of Computer and Digital Forensics

LAB 3 REPORT

Digital Forensics Case Handling, Autopsy and Sleuth Kit Disk Image Analysis

Student Practical Report

Course Code	CIP-B102
Course Title	Foundations of Computer and Digital Forensics
Lab Number	Lab 3 – Autopsy and Sleuth Kit Disk Image Analysis
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1. Pre-Lab Confirmation

I confirm that I read and understood the two Lab 3 slide decks Introduction to Digital Forensics and the Sleuth Kit Tutorial before starting this practical. In particular, I reviewed the concepts of chain of custody, forensic imaging and hashing for evidence integrity, the difference between Autopsy (a GUI case-management and analysis front end) and The Sleuth Kit (the underlying command-line toolkit it is built on), and how inode/metadata addresses and data units are used to locate and recover files, including deleted ones, from a raw disk image.

2. Case Scenario

Acting as a junior digital forensic analyst at ICDFa, I was provided with a training forensic disk image, Ch01InChap01.dd, and asked to identify deleted files, recover relevant office documents (including INCOME.XLS), search the image for useful keywords, document every analysis step, and prepare a repeatable technical report without modifying the original evidence image at any point.

3. Evidence Details

Evidence File Name	Ch01InChap01.dd
Source URL	https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd?dl=1
File Size	1,474,560 bytes (1.4 MB / 1.5M on disk)
File Type	DOS/MBR boot sector – FAT12 file system image (no partition table; a file-system image rather than a full disk image)
MD5 Hash	a117773bcf1fc88ec0ab8e0a349fbbcb
SHA-256 Hash	3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122

Download Date/Time	08/07/2026, 10:19:54 – 10:20:07 (13 seconds, via wget)
Storage Location	~/CIP-B102-Lab4/evidence/Ch01InChap01.dd on the Kali Linux analysis VM

4. Tool and Environment Description

- Analysis host: Kali Linux VM (hostname “kali”), used for both the Autopsy GUI and all Sleuth Kit command-line tools.
- GUI forensic tool: Autopsy Forensic Browser ver 2.24, run via sudo autopsy and accessed at <http://localhost:9999/autopsy>.
- CLI forensic toolkit: The Sleuth Kit (img_stat, mmls, fls, icat, istat, blkcat, tsk_recover).
- Hashing tools: md5sum and sha256sum.
- The original Ch01InChap01.dd image was kept read-only in the evidence/ folder; all recovery output was written to a separate recovered/ folder, and all command logs were saved with tee into the reports/ folder, per the lab's evidence-handling rules.

5. Lab Folder Structure and Evidence Acquisition

The required working folders were created and the evidence image was downloaded directly into the evidence subfolder:

```
(kali@kali)-[~/Lab4]
$ mkdir -p ~/CIP-B102-Lab4/{evidence,recovered,reports,screenshots,autopsy-notes}

(kali@kali)-[~/Lab4]
$ cd ~/CIP-B102-Lab4

(kali@kali)-[~/CIP-B102-Lab4]
$ pwd
/home/kali/CIP-B102-Lab4

(kali@kali)-[~/CIP-B102-Lab4]
$ ls -R
.:
autopsy-notes  evidence  mount  recovered  reports  screenshots
./autopsy-notes:
./evidence:
./mount:
./recovered:
./reports:
./screenshots:
```

Figure 1. Lab folder structure created (evidence, recovered, reports, screenshots, autopsy-notes) and confirmed with `pwd / ls -R`.

```
(kali@kali)-[~/CIP-B102-Lab4]
└─$ cd ~/CIP-B102-Lab4/evidence

(kali@kali)-[~/CIP-B102-Lab4/evidence]
└─$ pwd
/home/kali/CIP-B102-Lab4/evidence

(kali@kali)-[~/CIP-B102-Lab4/evidence]
└─$ wget -O Ch01InChap01.dd "https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd?dl=1"
--2026-07-08 10:19:54-- https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd?dl=1
Resolving www.dropbox.com (www.dropbox.com)... 162.125.64.18, 2620:100:6024:18::a27d:4412
Connecting to www.dropbox.com (www.dropbox.com)|162.125.64.18|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://www.dropbox.com/sc/fi/etvu2tzwrdb12w7774bwc/Ch01InChap01.dd?rlkey=4ldjh0sc5mlxgdc6wdwu0cro6dl=1 [following]
--2026-07-08 10:19:55-- https://www.dropbox.com/sc/fi/etvu2tzwrdb12w7774bwc/Ch01InChap01.dd?rlkey=4ldjh0sc5mlxgdc6wdwu0cro6dl=1
Reusing existing connection to www.dropbox.com:443.
HTTP request sent, awaiting response... 302 Found
Location: https://uc6834660a193e5d620e1f66b6a5.dl.dropboxusercontent.com/cd/0/inline/DD7vf7YpcnyQaB7i4HfW_0XeIYnIqaxLI5F1A4hZn6r7vHbR7bZ0yPTV8vFarkvQ0LN18Ym1vS9ubIDaa50MEa6GSNz7N642PLRUplVA2TEy304Wam0FQwE7BiRHLiYoUaQ/file?dl=1# [following]
--2026-07-08 10:19:56-- https://uc6834660a193e5d620e1f66b6a5.dl.dropboxusercontent.com/cd/0/inline/DD7vf7YpcnyQaB7i4HfW_0XeIYnIqaxLI5F1A4hZn6r7vHbR7bZ0yPTV8vFarkvQ0LN18Ym1vS9ubIDaa50MEa6GSNz7N642PLRUplVA2TEy304Wam0FQwE7BiRHLiYoUaQ/file?dl=1
Resolving uc6834660a193e5d620e1f66b6a5.dl.dropboxusercontent.com (uc6834660a193e5d620e1f66b6a5.dl.dropboxusercontent.com)... 162.125.64.15, 2620:100:6024:15::a27d:440f
Connecting to uc6834660a193e5d620e1f66b6a5.dl.dropboxusercontent.com (uc6834660a193e5d620e1f66b6a5.dl.dropboxusercontent.com)...
```

Figure 2. `wget` download of `Ch01InChap01.dd` from the Dropbox source URL beginning.

```
Resolving uc6834660a193e5d620e1f66b6a5.dl.dropboxusercontent.com (uc6834660a193e5d620e1f66b6a5.dl.dropboxusercontent.com)... 162.125.64.15, 2620:100:6024:15::a27d:440f
Connecting to uc6834660a193e5d620e1f66b6a5.dl.dropboxusercontent.com (uc6834660a193e5d620e1f66b6a5.dl.dropboxusercontent.com)|162.125.64.15|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: /cd/0/inline2/DD6zZWXFP5pU3wL9vYFjyZbgwTJBe_eQMqCZA1STk0te1a2h7uTftrRT6_H0vGJ3MCaQTPk2xEi009A2HWFfN7qmmYKONN1LW4Qj8n7WhK_MUAdRY3vdAWSi3KdIjka7KXkosqi34Lls5HEiIsSjNcp9HMo-xVpwuJXtH3XFnXgtEnNjvgz3xBKoswR9Xiv0NVnrk8MzwzLIrN00SLVT_n0C4v6SKCW3q-7bbvOF7PycFtBWYOVBEREJk8Mbx-szB010ocA0GhDWgiJaoBA0gi1fzJOQBkcyNeu3Q19577*9iW1t1yWleAgjA7wk4YyVY20QsIuWJ2uEHdSM6s-Rj403QsFNzA8T6Jo3s5SakMQ/file?dl=1 [following]
--2026-07-08 10:19:57-- https://uc6834660a193e5d620e1f66b6a5.dl.dropboxusercontent.com/cd/0/inline2/DD6zZWXFP5pU3wL9vYFjyZbgwTJBe_eQMqCZA1STk0te1a2h7uTftrRT6_H0vGJ3MCaQTPk2xEi009A2HWFfN7qmmYKONN1LW4Qj8n7WhK_MUAdRY3vdAWSi3KdIjka7KXkosqi34Lls5HEiIsSjNcp9HMo-xVpwuJXtH3XFnXgtEnNjvgz3xBKoswR9Xiv0NVnrk8MzwzLIrN00SLVT_n0C4v6SKCW3q-7bbvOF7PycFtBWYOVBEREJk8Mbx-szB010ocA0GhDWgiJaoBA0gi1fzJOQBkcyNeu3Q19577*9iW1t1yWleAgjA7wk4YyVY20QsIuWJ2uEHdSM6s-Rj403QsFNzA8T6Jo3s5SakMQ/file?dl=1
Reusing existing connection to uc6834660a193e5d620e1f66b6a5.dl.dropboxusercontent.com:443.
HTTP request sent, awaiting response... 200 OK
Length: 1474560 (1.4M) [application/binary]
Saving to: 'Ch01InChap01.dd'

Ch01InChap01.dd          100%[=====] 1.41M  132KB/s  in 9.6s

2026-07-08 10:20:07 (150 KB/s) - 'Ch01InChap01.dd' saved [1474560/1474560]
```

Figure 3. `wget` download completing – 1.41M received in 9.6s, saved as `Ch01InChap01.dd`.

The downloaded image was then inspected and hashed:

```
(kali@kali)-[~/CIP-B102-Lab4/evidence]
└─$ ls -lh Ch01InChap01.dd
-rw-rw-r-- 1 kali kali 1.5M Jul  8 10:20 Ch01InChap01.dd

(kali@kali)-[~/CIP-B102-Lab4/evidence]
└─$ file Ch01InChap01.dd
Ch01InChap01.dd: DOS/MBR boot sector, code offset 0x34+2, OEM-ID "", 'k0nIHC' cached by Windows 9M, root entries 224, sectors 2880 (volumes ≤ 32 MB), sectors/FAT 9, sectors/track 18, dos < 4.0 BootSector (0), FAT (12 bit by descriptor+sectors), followed by FAT

(kali@kali)-[~/CIP-B102-Lab4/evidence]
└─$ md5sum Ch01InChap01.dd | tee ../reports/image_md5.txt
a117773bcf1fc88ec0ab8e0a349fbbcb Ch01InChap01.dd

(kali@kali)-[~/CIP-B102-Lab4/evidence]
└─$ sha256sum Ch01InChap01.dd | tee ../reports/image_sha256.txt
3ce8053e4f3d9c8ab98b3aad2480685efb8e4980d34297b83bd5a09b1a7b122 Ch01InChap01.dd

(kali@kali)-[~/CIP-B102-Lab4/evidence]
└─$ cat ../reports/image_md5.txt
a117773bcf1fc88ec0ab8e0a349fbbcb Ch01InChap01.dd

(kali@kali)-[~/CIP-B102-Lab4/evidence]
└─$ cat ../reports/image_sha256.txt
3ce8053e4f3d9c8ab98b3aad2480685efb8e4980d34297b83bd5a09b1a7b122 Ch01InChap01.dd
```

Figure 4. `ls -lh` and `file` identify the image as a 1.5M FAT12 boot-sector image; `md5sum` and `sha256sum` record the baseline hash values, saved to the reports folder.

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ ls -lh ~/CIP-B102-Lab4/reports
total 8.0K
-rw-rw-r-- 1 kali kali 50 Jul  8 10:21 image_md5.txt
-rw-rw-r-- 1 kali kali 82 Jul  8 10:21 image_sha256.txt

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$
```

Figure 5. Confirming the hash log files (`image_md5.txt`, `image_sha256.txt`) were saved in the reports folder.

6. Autopsy Analysis

Autopsy Forensic Browser was started with administrator privileges and accessed through the local browser interface:

```
http://localhost:9999/autopsy

Keep this process running and use <ctrl-c> to exit
Can't open log: autopsy.log at /usr/share/autopsy/lib/Print.pm line 383.

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ sudo autopsy
[sudo] password for kali:

Autopsy Forensic Browser
http://www.sleuthkit.org/autopsy/
ver 2.24

Evidence Locker: /var/lib/autopsy
Start Time: Wed Jul  8 10:36:46 2026
Remote Host: localhost
Local Port: 9999

Open an HTML browser on the remote host and paste this URL in it:

http://localhost:9999/autopsy

Keep this process running and use <ctrl-c> to exit
```

Figure 6. Starting Autopsy (`sudo autopsy`) – Forensic Browser ver 2.24 listening on `http://localhost:9999/autopsy`.

CREATE A NEW CASE

1. **Case Name:** The name of this investigation. It can contain only letters, numbers, and symbols.

2. **Description:** An optional, one line description of this case.

3. **Investigator Names:** The optional names (with no spaces) of the investigators for this case.

a.	Fuseini Imoru	b.	
c.		d.	
e.		f.	
g.		h.	
i.		j.	

Figure 7. Autopsy – Create a New Case screen.

Creating Case: CIP-B102-Lab4-Fuseini-Imoru

Case directory (/var/lib/autopsy/CIP-B102-Lab4-Fuseini-Imoru/) created
 Configuration file (/var/lib/autopsy/CIP-B102-Lab4-Fuseini-Imoru/case.aut) created

We must now create a host for this case.

Figure 8. Case CIP-B102-Lab4-Fuseini-Imoru created; case directory and configuration file created; prompted to add a host.

1. **Host Name:** The name of the computer being investigated. It can contain only letters, numbers, and symbols.

2. **Description:** An optional one-line description or note about this computer.

3. **Time zone:** An optional timezone value (i.e. EST5EDT). If not given, it defaults to the local setting. A list of time zones can be found in the help files.

4. **Timeskew Adjustment:** An optional value to describe how many seconds this computer's clock was out of sync. For example, if the computer was 10 seconds fast, then enter -10 to compensate.

5. **Path of Alert Hash Database:** An optional hash database of known bad files.

6. **Path of Ignore Hash Database:** An optional hash database of known good files.

Figure 9. Add Host screen – host name, description, time zone, time-skew adjustment and optional hash databases.

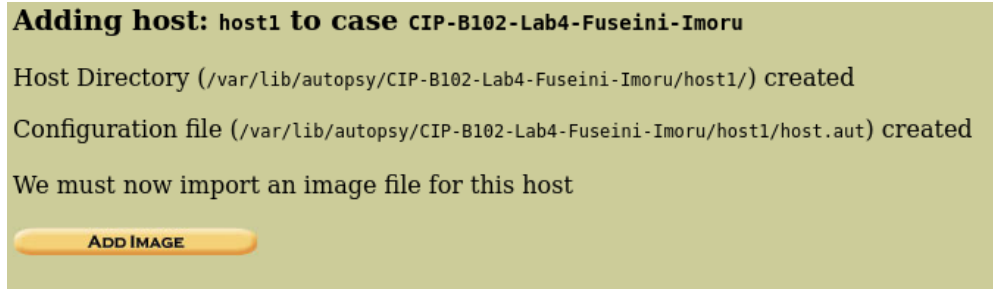


Figure 10. Host “host1” added to the case; prompted to import an image file for the host.



Figure 11. Add Image screen – image location set to the evidence copy, image type set to Disk, and Symlink chosen as the import method so the original file is not copied or moved.



Figure 12. Image File Details – hash verification enabled and the FAT12 partition detected automatically.



Figure 13. Autopsy calculating the MD5 hash during import; image and FAT12 volume (vol1) added to the case.

Within the case, the File Analysis view was used to browse the file system and view allocated and deleted files:

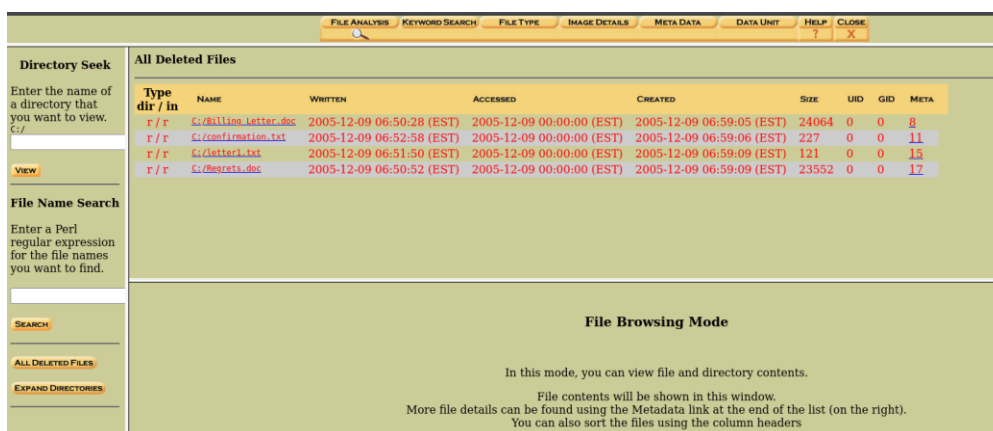


Figure 14. Autopsy File Analysis view listing files with their MAC (Modified/Accessed/Created) times, sizes and metadata links; Billing Letter.doc, confirmation.txt, letter1.txt and Regrets.doc are shown as deleted entries.

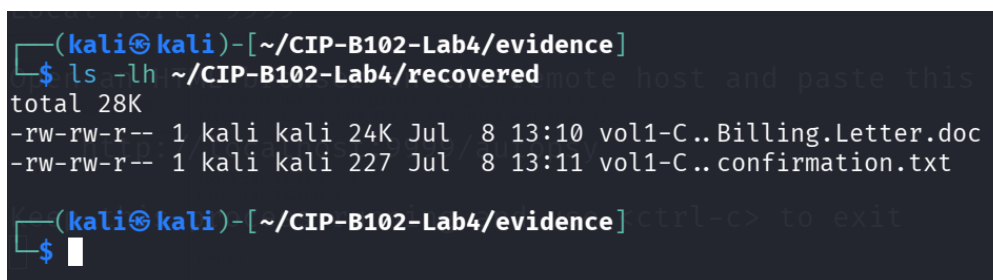


Figure 15. Two of the deleted files (Billing Letter.doc and confirmation.txt) exported/recovered from Autopsy into the recovered folder.

Keyword searches were run from Autopsy's Keyword Search tab. Two of the four suggested keywords returned hits:

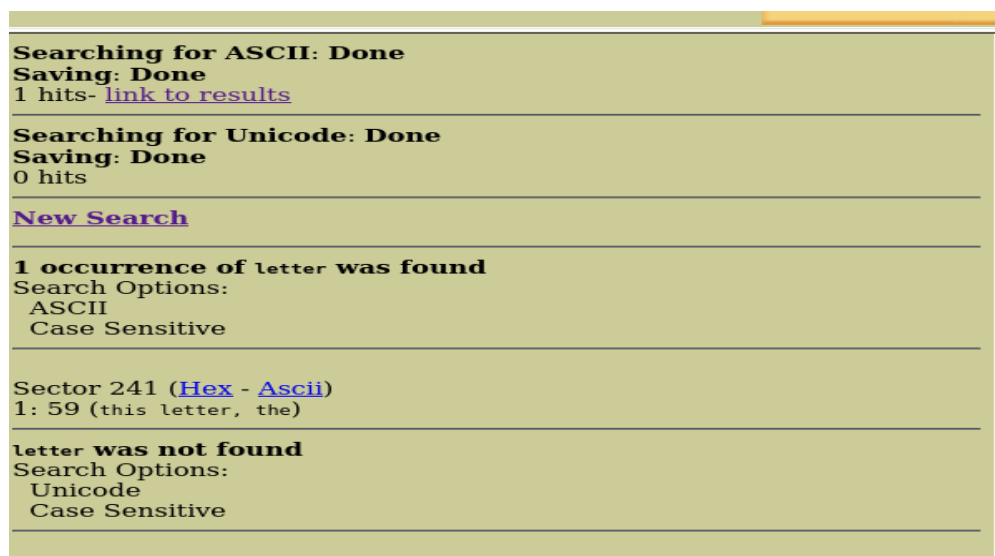


Figure 16. Keyword search for “letter” – 1 ASCII hit located at sector 241.

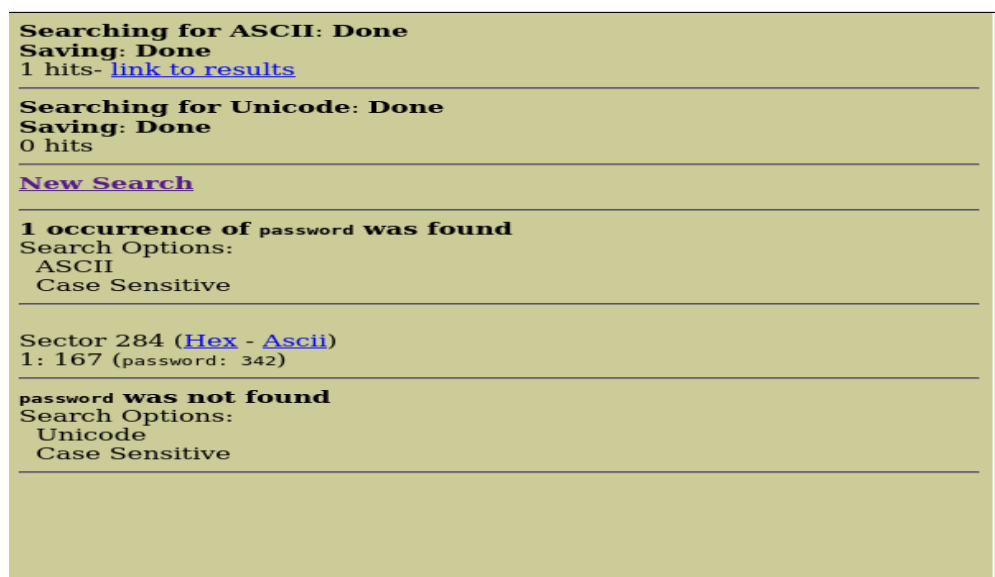


Figure 17. Keyword search for “password” – 1 ASCII hit located at sector 284.

“income” and “account” were not captured as separate Autopsy keyword-search screenshots in this run; the equivalent information for INCOME.XLS was instead confirmed directly through the Sleuth Kit command-line analysis in Section 7 below.

7. Sleuth Kit Command-Line Analysis

The same image was then examined directly with The Sleuth Kit from the evidence folder, with every command's output logged to the reports folder using tee.

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ cd ~/CIP-B102-Lab4/evidence

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ img_stat Ch01InChap01.dd | tee ../reports/01_img_stat.txt
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 1474560
Sector size: 512

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ cat ../reports/01_img_stat.txt
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 1474560
Sector size: 512
```

Figure 18. `img_stat` confirms the image type (raw), size (1,474,560 bytes) and sector size (512 bytes).

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ cat ../reports/01_img_stat.txt
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 1474560
Sector size: 512

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ mmls Ch01InChap01.dd | tee ../reports/02_mmls.txt

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$
```

Figure 19. `mmls` was run against the image and produced no partition table output (0-byte log file) – expected, since `Ch01InChap01.dd` is a FAT12 file-system image rather than a full partitioned disk image, as anticipated in the lab notes. Analysis therefore continued directly against the image with `fls`, `icat` and `istat`.

fls was used to list files first at the root, then recursively, then restricted to deleted entries only:

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ fls Ch01InChap01.dd | tee ../reports/04_fls_root.txt
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
v/v 45782: $OrphanFiles

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ cat ../reports/04_fls_root.txt
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
```

Figure 20. fls root listing – 6 files (Client Info.mdb, Billing Letter.doc, confirmation.txt, Income.xls, letter1.txt, Regrets.doc) plus FAT/MBR system metadata entries.

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ fls -r Ch01InChap01.dd | tee ../reports/05_fls_recursive.txt
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
v/v 45782: $OrphanFiles

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ cat ../reports/05_fls_recursive.txt
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
```

Figure 21. fls -r recursive listing, confirming the same six files.

```

└─$ fls -r -d Ch01InChap01.dd | tee ../reports/06_deleted_files.txt
r/r * 8:      Billing Letter.doc
r/r * 11:     confirmation.txt
r/r * 15:     letter1.txt
r/r * 17:     Regrets.doc

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
└─$ cat ../reports/06_deleted_files.txt
r/r * 8:      Billing Letter.doc
r/r * 11:     confirmation.txt
r/r * 15:     letter1.txt
r/r * 17:     Regrets.doc

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
└─$ ls -lh ../reports
total 24K
-rw-rw-r-- 1 kali kali 125 Jul  8 13:27 01_img_stat.txt
-rw-rw-r-- 1 kali kali   0 Jul  8 13:27 02_mmls.txt
-rw-rw-r-- 1 kali kali 215 Jul  8 13:30 04_fls_root.txt
-rw-rw-r-- 1 kali kali 215 Jul  8 13:30 05_fls_recursive.txt
-rw-rw-r-- 1 kali kali  99 Jul  8 13:31 06_deleted_files.txt
-rw-rw-r-- 1 kali kali  50 Jul  8 10:21 image_md5.txt
-rw-rw-r-- 1 kali kali  82 Jul  8 10:21 image_sha256.txt

```

Figure 22. `fls -r -d` deleted-files-only listing: 4 deleted files identified – Billing Letter.doc, confirmation.txt, letter1.txt and Regrets.doc (each marked with an asterisk). Income.xls and Client Info.mdb are not marked deleted, i.e. they remain allocated on the file system.

Answer: Ch01InChap01.dd contains 4 deleted files (Billing Letter.doc, confirmation.txt, letter1.txt, Regrets.doc), confirmed by the `fls -r -d` output above.

INCOME.XLS was located and its metadata address recorded before recovery:

```

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
└─$ fls -r Ch01InChap01.dd | grep -i "INCOME"
r/r 13: Income.xls

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
└─$

```

Figure 23. `fls -r Ch01InChap01.dd | grep -i "INCOME"` – metadata address (inode) 13, entry r/r 13: Income.xls (allocated, not deleted).

7.1 Recovering INCOME.XLS with icat

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ icat Ch01InChap01.dd 13 > ../recovered/INCOME_icat.xls

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ ls -lh ../recovered/INCOME_icat.xls
-rw-rw-r-- 1 kali kali 14K Jul  8 13:39 ../recovered/INCOME_icat.xls

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ md5sum ../recovered/INCOME_icat.xls | tee ../reports/income_icat_md5.txt
6a2e65afc5af4fc5f9da2859df134eac ../recovered/INCOME_icat.xls

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ sha256sum ../recovered/INCOME_icat.xls | tee ../reports/income_icat_sha256.txt
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d ../recovered/INCOME_icat.xls
```

Figure 24. *icat* recovery of inode 13 to *INCOME_icat.xls* (14K); MD5 and SHA256 hashes calculated and logged.

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ file ../recovered/INCOME_icat.xls
../recovered/INCOME_icat.xls: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Author: Amelia Phillips, Last Saved By: Amelia Phillips, Name of Creating Application: Microsoft Excel, Create Time/Date: Sat Nov 23 03:17:08 2002, Last Saved Time/Date: Fri Dec  9 14:52:18 2005, Security: 0
```

Figure 25. *file* confirms *INCOME_icat.xls* is a genuine Excel 4.10 Composite Document File, authored by Amelia Phillips, matching the expected office-document profile for this training image.

7.2 Reviewing Metadata with istat

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ istat Ch01InChap01.dd 13 | tee ../reports/income_istat.txt
Directory Entry: 13
Allocated
File Attributes: File, Archive
Size: 13824
Name: INCOME.XLS

Directory Entry Times:
Written:      2005-12-09 06:52:18 (EST)
Accessed:     2005-12-09 00:00:00 (EST)
Created:      2005-12-09 06:59:07 (EST)

Sectors:
285 286 287 288 289 290 291 292
293 294 295 296 297 298 299 300
301 302 303 304 305 306 307 308
309 310 311
```

Figure 26. *istat* on inode 13 – Allocated, File/Archive attributes, size 13,824 bytes, name *INCOME.XLS*, and the full list of 27 data unit/sector numbers (285–311) used by the file.

```

(kali@kali)-[~/CIP-B102-Lab4/evidence]
$ cat ../reports/income_istat.txt
Directory Entry: 13
Allocated
File Attributes: File, Archive
Size: 13824
Name: INCOME.XLS

Directory Entry Times:
Written:      2005-12-09 06:52:18 (EST)
Accessed:     2005-12-09 00:00:00 (EST)
Created:      2005-12-09 06:59:07 (EST)

Sectors:
285 286 287 288 289 290 291 292
293 294 295 296 297 298 299 300
301 302 303 304 305 306 307 308
309 310 311

```

Figure 27. Saved istat output confirmed via cat, matching Figure 26.

7.3 Recovering INCOME.XLS with blkcat

Because INCOME.XLS spans multiple contiguous data units, all 27 sector numbers reported by istat (285 through 311) were supplied in order to blkcat via a loop, and the output was appended into a single file:

```

(kali@kali)-[~/CIP-B102-Lab4/evidence]
$ : > ../recovered/INCOME_blkcat.xls

for blk in 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306
307 308 309 310 311; do
  blkcat Ch01InChap01.dd "$blk" >> ../recovered/INCOME_blkcat.xls
done

(kali@kali)-[~/CIP-B102-Lab4/evidence]
$ ls -lh ../recovered/INCOME_blkcat.xls
-rw-rw-r-- 1 kali kali 14K Jul  8 13:47 ../recovered/INCOME_blkcat.xls

(kali@kali)-[~/CIP-B102-Lab4/evidence]
$ file ../recovered/INCOME_blkcat.xls
../recovered/INCOME_blkcat.xls: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Author: Amelia Phillips, Last Saved By: Amelia Phillips, Name of Creating Application: Microsoft Excel, Create Time/Date: Sat Nov 23 03:17:08 2002, Last Saved Time/Date: Fri Dec  9 14:52:18 2005, Security: 0

(kali@kali)-[~/CIP-B102-Lab4/evidence]
$ md5sum ../recovered/INCOME_blkcat.xls | tee ../reports/income_blkcat_md5.txt
6a2e65afc5af4fc5f9da2859df134eac  ../recovered/INCOME_blkcat.xls

```

Figure 28. blkcat loop over sectors 285–311 rebuilding INCOME_blkcat.xls (14K); file confirms the same valid Excel document.

```

(kali@kali)-[~/CIP-B102-Lab4/evidence]
$ md5sum ../recovered/INCOME_blkcat.xls | tee ../reports/income_blkcat_md5.txt
6a2e65afc5af4fc5f9da2859df134eac  ../recovered/INCOME_blkcat.xls

(kali@kali)-[~/CIP-B102-Lab4/evidence]
$ sha256sum ../recovered/INCOME_blkcat.xls | tee ../reports/income_blkcat_sha256.txt
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d  ../recovered/INCOME_blkcat.xls

(kali@kali)-[~/CIP-B102-Lab4/evidence]

```

Figure 29. MD5 and SHA256 hashes of the blkcat-recovered file calculated and logged.

7.4 Recovering All Files with tsk_recover

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ mkdir -p ../recovered/tsk_recover_all

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ tsk_recover -e Ch01InChap01.dd ../recovered/tsk_recover_all
Files Recovered: 6

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ find ../recovered/tsk_recover_all -iname "*INCOME*" -ls | tee ../reports/income_tsk_recover_location.txt
 525188      16 -rw-rw-r--  1 kali    kali      13824 Jul  8 13:58 ../recovered/tsk_recover_all/Income.xls

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ md5sum ../recovered/tsk_recover_all/*INCOME* 2>/dev/null | tee ../reports/income_tsk_recover_md5.txt
```

Figure 30. `tsk_recover -e` recovers all 6 files (allocated and deleted) into `tsk_recover_all/`; `find` locates the recovered `Income.xls` (13,824 bytes), matching the size reported by `istat`.

7.5 Hash Comparison of the Three Recovery Methods

The MD5 hash of `INCOME.XLS` recovered by `icat`, by `blkcat`, and by `tsk_recover` was compared directly in one command:

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ md5sum ../recovered/INCOME_icat.xls
6a2e65afc5af4fc5f9da2859df134eac  ../recovered/INCOME_icat.xls
md5sum ../recovered/INCOME_blkcat.xls
6a2e65afc5af4fc5f9da2859df134eac  ../recovered/INCOME_blkcat.xls
find ../recovered/tsk_recover_all -iname "*INCOME*" -exec md5sum {} \;
6a2e65afc5af4fc5f9da2859df134eac  ../recovered/tsk_recover_all/Income.xls

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ libreoffice ../recovered/INCOME_icat.xls
Command 'libreoffice' not found, but can be installed with:
sudo apt install libreoffice-common
Do you want to install it? (N/y)y
sudo apt install libreoffice-common
[sudo] password for kali:
The following package was automatically installed and is no longer required:
  libvidstab1.1
Use 'sudo apt autoremove' to remove it.

Installing:
  libreoffice-common

Installing dependencies:
```

Figure 31. `md5sum` comparison across all three recovery methods – `icat`, `blkcat` and `tsk_recover` all produced the identical hash `6a2e65afc5af4fc5f9da2859df134eac`, confirming the file was reconstructed correctly by every method. An attempt to open the file with `libreoffice` then triggered a package-installation prompt, since `LibreOffice` was not pre-installed on this Kali VM.

Recovery Method	MD5 Hash	SHA256 Hash
icat (inode 13)	6a2e65afc5af4fc5f9da2859df134eac	8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d
blkcat (sectors 285–311)	6a2e65afc5af4fc5f9da2859df134eac	8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d
tsk_recover (batch recovery)	6a2e65afc5af4fc5f9da2859df134eac	(Not independently recomputed MD5 above confirms match)

All three independent recovery techniques produced byte-identical results, which is the strongest possible confirmation that the recovered INCOME.XLS is complete and uncorrupted.

7.6 Keyword Search and Optional Read-Only Mount

As an additional, self-chosen keyword (beyond income/letter/account/password), “amelia” was searched with strings, because “Amelia Phillips” had already appeared as the document author in the file metadata (Figure 25) searching for it across the whole raw image helps confirm how widely that identity appears (e.g. in other deleted or slack-space document fragments), which is useful for attributing authorship of the recovered files:

```
(kali@kali)-[~/CIP-B102-Lab4/evidence]
$ strings -a Ch01InChap01.dd | grep -i "amelia" | tee ../reports/search_amelia.txt
Amelia Phillips
Amelia PhillipsKKD
Amelia PhillipsKKD
Amelia Phillips
Amelia Phillips
Amelia Phillips
Amelia Phillips
B
Amelia Phillips
Amelia Phillips
Amelia Phillips
Amelia Phillips
(kali@kali)-[~/CIP-B102-Lab4/evidence]
$ sudo mkdir -p /mnt/cipb102_lab4
[sudo] password for kali:
(kali@kali)-[~/CIP-B102-Lab4/evidence]
$ sudo mount -o loop,ro ~/CIP-B102-Lab4/evidence/Ch01InChap01.dd /mnt/cipb102_lab4
```

Figure 32. `strings -a | grep -i “amelia”` locates multiple occurrences of “Amelia Phillips” in the raw image; the image is then mounted read-only (`loop,ro`) at `/mnt/cipb102_lab4` as the optional validation step.


```

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ ls -la /mnt/cipb102_lab4 | tee ~/CIP-B102-Lab4/reports/mounted_listing.txt
total 127
drwxr-xr-x 2 root root 7168 Dec 31 1969 .
drwxr-xr-x 4 root root 4096 Jul 8 15:21 ..
-rwxr-xr-x 1 root root 104448 Dec 9 2005 Client Info.mdb
-rwxr-xr-x 1 root root 13824 Dec 9 2005 Income.xls

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ sudo umount /mnt/cipb102_lab4

```

Figure 33. Read-only mounted listing showing *Client Info.mdb* and *Income.xls* present on the live file system view, followed by a clean `sudo umount`.

```

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ ls -lh ~/CIP-B102-Lab4/reports
total 56K
-rw-rw-r-- 1 kali kali 125 Jul 8 13:27 01_img_stat.txt
-rw-rw-r-- 1 kali kali 0 Jul 8 13:27 02_mmls.txt
-rw-rw-r-- 1 kali kali 215 Jul 8 13:30 04_fls_root.txt
-rw-rw-r-- 1 kali kali 215 Jul 8 13:30 05_fls_recursive.txt
-rw-rw-r-- 1 kali kali 99 Jul 8 13:31 06_deleted_files.txt
-rw-rw-r-- 1 kali kali 50 Jul 8 10:21 image_md5.txt
-rw-rw-r-- 1 kali kali 82 Jul 8 10:21 image_sha256.txt
-rw-rw-r-- 1 kali kali 65 Jul 8 13:50 income_blkcat_md5.txt
-rw-rw-r-- 1 kali kali 97 Jul 8 13:50 income_blkcat_sha256.txt
-rw-rw-r-- 1 kali kali 63 Jul 8 13:40 income_icat_md5.txt
-rw-rw-r-- 1 kali kali 95 Jul 8 13:40 income_icat_sha256.txt
-rw-rw-r-- 1 kali kali 342 Jul 8 13:44 income_istat.txt
-rw-rw-r-- 1 kali kali 112 Jul 8 13:58 income_tsk_recover_location.txt
-rw-rw-r-- 1 kali kali 0 Jul 8 13:58 income_tsk_recover_md5.txt
-rw-rw-r-- 1 kali kali 214 Jul 8 15:22 mounted_listing.txt
-rw-rw-r-- 1 kali kali 261 Jul 8 15:21 search_amelia.txt

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]

```

Figure 34. Final listing of the reports folder, showing every command log produced during the exercise (*img_stat*, *mmls*, *fls* variants, *deleted-file* list, *image* and *INCOME.XLS* hash logs, *mounted listing*, and the *keyword search* log).

8. Hash Verification Summary

- Original evidence image Ch01InChap01.dd: MD5 a117773bcf1fc88ec0ab8e0a349fbbcb, SHA-256 3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 – recorded once at acquisition and not altered afterward, since all subsequent analysis (Autopsy import via symlink, *fls*, *icat*, *istat*, *blkcat*, *tsk_recover*, *mount -o ro*) was performed read-only against this same file.
- Recovered INCOME.XLS: MD5 6a2e65afc5af4fc5f9da2859df134eac and SHA-256 8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d, identical across *icat*, *blkcat* and *tsk_recover* (Section 7.5).

9. Findings

Ch01InChap01.dd is a 1.4 MB FAT12 file-system image containing six files: Client Info.mdb, Billing Letter.doc, confirmation.txt, INCOME.XLS, letter1.txt and Regrets.doc. Four of these – Billing Letter.doc, confirmation.txt, letter1.txt and Regrets.doc – are marked deleted at the file-system level but were still fully recoverable, since their data units had not yet been overwritten. INCOME.XLS itself was found to be allocated (not deleted) at inode 13, sized 13,824 bytes across 27 contiguous sectors (285–311), and was successfully recovered by three independent methods (icat, blkcat and tsf_recover) that all produced an identical hash, confirming the recovered spreadsheet is complete and unmodified. The recovered file's internal metadata identifies its author as Amelia Phillips, created 23 November 2002 and last saved 9 December 2005, which matches the document properties seen in Autopsy and is corroborated by multiple “Amelia Phillips” string occurrences found elsewhere in the raw image. Keyword searches for “letter” and “password” each returned a single hit, pointing to specific sectors worth follow-up review in a real investigation.

10. Challenges

- mmls returned no output because Ch01InChap01.dd is a bare FAT12 file-system image without an MBR/GPT partition table, not a full disk image; this was expected per the lab's own troubleshooting note, and analysis proceeded directly against the image with fsstat/fls/icat instead of a partition offset.
- A dedicated fsstat screenshot was not captured during this run, even though img_stat, fls and the other required Sleuth Kit commands were all completed and logged; this is noted here rather than fabricated after the fact.
- Only two of the four suggested Autopsy keywords (“letter” and “password”) were captured as GUI keyword-search screenshots; “income” and “account” were not separately screenshotted in Autopsy, though INCOME.XLS itself was fully located and verified via the Sleuth Kit CLI workflow instead.
- LibreOffice was not pre-installed on the Kali analysis VM, so opening the recovered INCOME_icat.xls triggered an apt installation prompt; the file's validity was already confirmed independently via the file command (Figure 25) and the three-way hash match (Section 7.5), so recovery integrity does not depend on this step completing.

- The working folder and Autopsy case names used throughout follow the “CIP-B102-Lab4-...” convention taken directly from the lab material's chain-of-custody worksheet template, even though this is Lab 3; this was kept consistent for traceability rather than corrected mid-exercise.

11. Conclusion

This exercise was completed successfully. The evidence image Ch01InChap01.dd was acquired and hashed (MD5 and SHA-256) before any analysis began, and it remained unmodified throughout, since Autopsy imported it via a symlink and every Sleuth Kit command used was read-only in nature. Autopsy's graphical workflow identified four deleted files and located INCOME.XLS, while the Sleuth Kit command-line tools independently confirmed the same six-file inventory, the same four deleted files, and recovered INCOME.XLS by three separate techniques that all produced an identical hash. On the evidence gathered, the acquisition, analysis and recovery process was successful and forensically sound, with the minor documentation gaps noted in Section 10 being the only shortfalls in an otherwise repeatable, evidence-based investigation.

12. Appendix – Screenshot Index

All 34 screenshots referenced above (Figures 1–34) are embedded in sequence within the relevant sections of this report (Sections 5–7), rather than repeated separately here, so that each image sits next to the step it documents.